

Biztel AI Supervisor

Engine Cleaning — Live Demo Walkthrough

Real-time AI vision that tracks surface cleaning coverage, verifies completion automatically, and eliminates manual inspection.

EDGE AI

REAL-TIME QA

INSPECTION

What is Biztel AI Supervisor?

Biztel AI Supervisor is an edge-AI quality assurance system that mounts a camera above each manufacturing station and tracks worker activity in real time. For engine cleaning, it uses computer vision to map exactly which surface areas have been cleaned and which have not — providing live coverage feedback to the operator and automatically logging a **PASS** or **FAIL** at cycle end.

There is no wearable hardware, no barcodes, and no operator input required. The system runs entirely on a local edge device, with no cloud dependency during production.

How the colour overlay works

-  **Yellow — Area has not yet been cleaned. AI is watching.**
-  **Green — Area confirmed cleaned. AI has verified contact.**
-  **Blue — The cleaning cloth, tracked by AI in real time.**

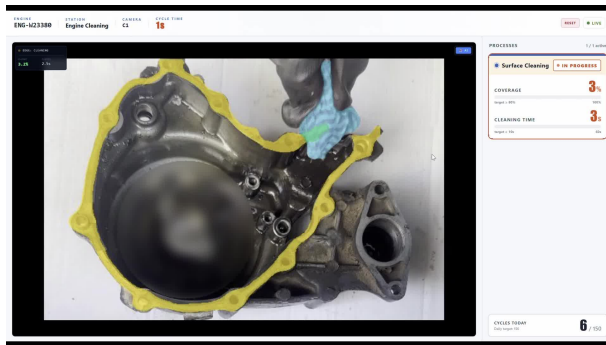
What the dashboard shows

Field	What it means
ENGINE	Unique part ID — auto-detected at cycle start
STATION	Workstation name (Engine Cleaning, Gasket Check, etc.)
CYCLE TIME	Seconds elapsed for this part — resets on new part
COVERAGE %	Percentage of the defined surface area verified clean
CLEANING TIME	Time spent actively cleaning (vs. idle). Target shown.
Status: OK / IN PROGRESS / FAIL	Auto-issued verdict. No human sign-off needed.
CYCLES TODAY	Parts completed at this station vs. daily target

A Single Cleaning Cycle — Step by Step

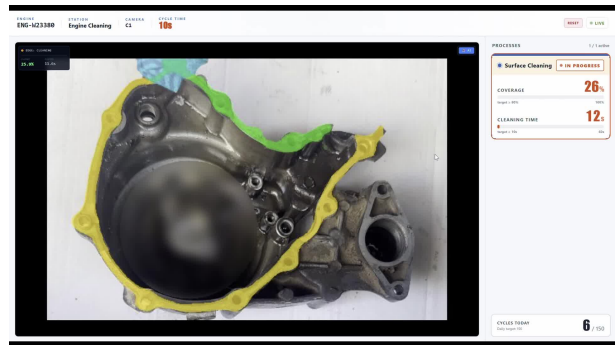
The sequence below shows one complete engine cleaning cycle (ENG-W23380) from the demo video. The full cycle took **31 seconds** and achieved **99% coverage** — well above the 80% minimum.

① Cycle underway



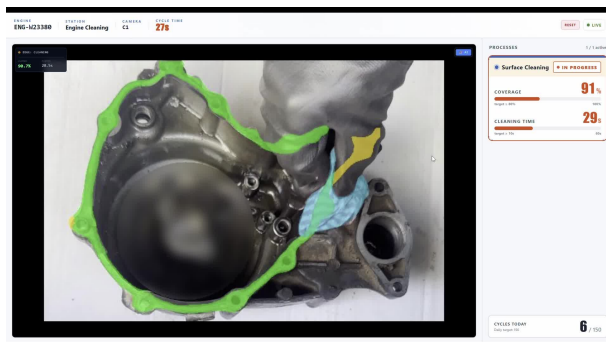
Engine part detected. Coverage at 3% — almost entirely yellow. Cleaning has just begun.

② AI tracks in real time



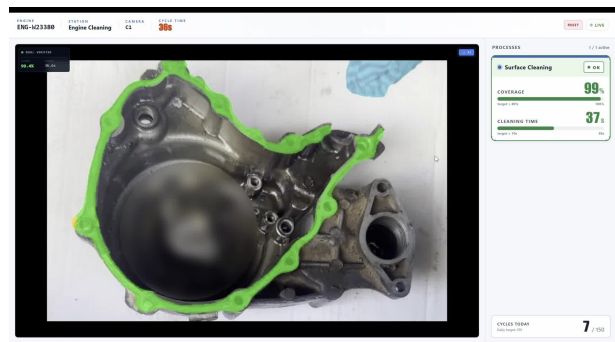
As the operator cleans, yellow transitions to green. Coverage shown live. No manual logging.

③ Auto-verified



Coverage hits 99% at 31 seconds — above the 80% target. Status flips to OK automatically.

④ Next cycle begins



AI instantly resets for the next part. A new engine ID is detected and tracking restarts.

Key Performance Metrics

31_s

Avg. cycle time (target ≤ 60s)

99%

Surface coverage achieved (target
≥ 80%)

150/day

Daily cycle target per station

Why this matters

- **Eliminates manual inspection** — no quality supervisor needed at this station. Each OK is issued by AI in under a second.
- **Prevents defect escape** — insufficiently cleaned engine parts cause downstream failures. The 80% coverage floor is enforced on every single part.
- **Creates a full audit trail** — every cycle is logged with engine ID, coverage %, cleaning time, and timestamp. Fully traceable.
- **Edge-native, no cloud dependency** — runs on a local device at the station. Works in facilities with no internet connectivity.
- **Works with existing operators** — no retraining, no wearables, no workflow change. The system adapts to the operator, not the other way around.
- **Scales across stations** — the same software module applies to gasket cleaning, surface prep, and any coverage-based quality check.

This demo covers one station (Engine Cleaning). Biztel AI Supervisor is already running across multiple process types. Contact us to schedule a live factory walkthrough.